Do Teacher–Student Alignment and Gaussian KL Improve Short and Noisy Vietnamese Speaker Verification?

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ABSTRACT

Short and noisy probes degrade speaker verification through limited speaker evidence and mismatch with cleaner enrollment speech. We evaluate ECAPA-TDNN adaptation for Vietnamese using four historical systems: an unadapted pretrained encoder, AAM-only adaptation, vector alignment with an exponential moving average (EMA) teacher, and a full model adding symmetric batch-level diagonal Gaussian-KL alignment. Adapted systems use 3,371 training speakers. A frozen 5,000-trial sample of speaker-disjoint VieSpeaker-H is evaluated under 27 combinations of enrollment duration (2, 3, 4 s), probe duration (1.0, 1.25, 1.5 s), and clean, 10-dB, or 0-dB probes. Models share scorable trial IDs within each experiment family and condition. At 3-s enrollment and 1.25-s probes, AAM-only reduces pretrained EER from 20.031% to 12.700% clean, 24.037% to 15.436% at 10 dB, and 32.783% to 23.817% at 0 dB. Within the historical objective family, AAM-only has the lowest point estimate in 24 conditions; vector-EMA leads in three and Full in none. The saved paired intervals do not establish an auxiliary gain. C/D retain teacher enrollment and student probe encoders at verification. A separate exploratory pooling study uses 3,359 training speakers: global/local summaries and a batch-relative length gate show condition-specific changes without a consistent aggregate gain. Single training runs, training-provenance differences, and duration-dependent exclusions limit causal conclusions.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**.

KEYWORDS

automatic speaker verification, Vietnamese speech, short utterances, noise robustness, ECAPA-TDNN, additive angular margin, teacher–student learning, EMA, embedding alignment, Gaussian KL divergence, equal error rate

1 INTRODUCTION

Speaker verification compares a probe recording to an enrolled voice to test a claimed identity [1]. Short utterances carry less speaker information [2]; noise further degrades short-probe verification when enrollment is cleaner [3]. We study text-independent probes of 1.0–1.5 s. A personalized assistant motivates this setting, but we do not evaluate wake-word detection or an integrated assistant [4].

Vietnamese speaker verification under short and noisy probe conditions has received limited attention. Although Vietnamese speaker datasets and verification systems have recently emerged [5–7], differing datasets, partitions, durations, noise conditions, and training methods make it difficult to know whether more complex training reliably improves upon a strong AAM-only baseline.

We compare a pretrained ECAPA-TDNN [8], AAM-only adaptation [9], EMA teacher–student vector alignment [10], and an additional batch-level Gaussian-KL loss. All four use frozen trials across 27 duration–noise conditions. We ask whether either auxiliary objective adds consistent gains beyond supervised adaptation. A secondary experiment explores original, global/local, and batch-relative-length pooling. We distinguish these architecture variants from Systems A/B/C/D and retain their separate training provenance.

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2 RELATED WORK

Early i-vector factor analysis [11] preceded neural speaker embeddings. X-vectors use a TDNN and statistics pooling for discriminative embeddings [12]. ECAPA-TDNN adds channel attention and multi-layer aggregation [8]; CAM++ is another efficient backbone [13]. Additive angular margins originated in face recognition [14] and improve speaker embeddings [9], though metric learning remains competitive [15]. Short and noisy speech motivated length/noise-aware training [3], long-to-short teacher–student alignment [2], and diffusion-based feature extension [16]. We use an EMA teacher, following Mean Teacher [10], but match batch-level embedding moments rather than the classification outputs studied by Jung et al. [2].

Vietnamese verification datasets include VietSV [5], VoxVietnam [6], and VieSpeaker [7]; VoxCeleb1 and VoxCeleb2 are larger in-the-wild reference corpora [17, 18]. VieSpeaker supplies our training data and a held-out speaker-disjoint protocol. Our comparison holds data, trial IDs, durations, and noise conditions fixed across models within each condition.

3 SYSTEM

All systems use the same pretrained ECAPA-TDNN backbone [8]. A is the unadapted checkpoint; B adapts its encoder using AAM-Softmax [9]. C adds an EMA teacher receiving a longer unaugmented crop, while its student receives a short augmented crop from the same speaker. D also aligns the student and teacher embedding distributions. At step k, the teacher update and vector loss are

$$\theta_t^{(k)} = \alpha\theta_t^{(k-1)} + (1-\alpha)\theta_s^{(k)}, \tag{1}$$

$$L_{\text{vec}} = \frac{1}{N}\sum_{i=1}^{N}(1 - \mathbf{z}_{t,i}^{\top}\mathbf{z}_{s,i}), \tag{2}$$

where $\mathbf{z}$ are unit-normalized embeddings of paired crops and N is the number of pairs. The teacher receives no gradients from the student loss. EMA smooths the reference across updates, while L_{vec} acts on corresponding long and short utterance crops rather than on aggregate speaker classes. The student also receives an AAM classification loss using the training-speaker labels.

For branch $b \in \{s, t\}$, the batch moments are

$$\boldsymbol{\mu}_b = N^{-1}\sum_{i=1}^{N}\mathbf{z}_{b,i}, \tag{3}$$

$$\boldsymbol{\sigma}_b^2 = N^{-1}\sum_{i=1}^{N}(\mathbf{z}_{b,i} - \boldsymbol{\mu}_b)^2 + \varepsilon. \tag{4}$$

The diagonal Gaussian $q_b = \mathcal{N}(\boldsymbol{\mu}_b, \operatorname{diag}\boldsymbol{\sigma}_b^2)$ describes minibatch moments, not per-utterance uncertainty. In particular, each batch mixes multiple identities, so the variance is not a calibrated measure of an individual utterance's uncertainty. D adds

$$L_{\text{KL}} = \tfrac{1}{2}[D_{\text{KL}}(q_s\|q_t) + D_{\text{KL}}(q_t\|q_s)]. \tag{5}$$

The objective is $L = L_{\text{AAM}} + \lambda_{\text{vec}}L_{\text{vec}} + \lambda_{\text{KL}}L_{\text{KL}}$, with zero auxiliary coefficients in B and $\lambda_{\text{KL}} = 0$ in C. For C, $\lambda_{\text{vec}} = 1$; for D, $\lambda_{\text{vec}} = 1$ and $\lambda_{\text{KL}} = 0.01$. The Gaussian loss matches batch-level marginal moments and does not explicitly enforce class-conditional agreement. Consequently, any effect on speaker discrimination must be established empirically rather than assumed from a reduced KL value.

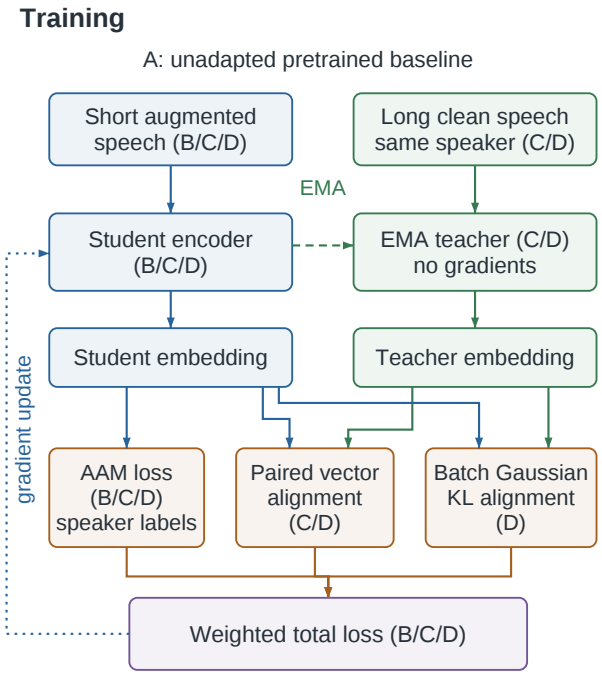


Figure 1: Training. Solid arrows carry forward computations; dotted blue arrows indicate student gradient updates, and dashed green arrows indicate EMA parameter updates. The teacher receives no gradients. AAM uses speaker labels, vector alignment uses paired embeddings, and KL uses minibatch moments. Only active losses enter the weighted total.

Figure 1 separates forward computation from parameter updates. The AAM head and auxiliary losses are inactive at verification. A uses the pretrained encoder for both enrollment and probes; B uses its student for both. The recorded C/D evaluation uses the EMA teacher for enrollment and the student for probes, so the teacher is retained (Figure 2). Unit-normalized embeddings are scored by cosine similarity; a score at least τ accepts the claim. EER varies τ to equalize false-accept and false-reject rates. Thus B versus C/D changes both training and inference roles, rather than isolating an auxiliary loss.

4 EXPERIMENTAL SETUP

4.1 Data and Preprocessing

The data are drawn from VieSpeaker [7]. VieSpeaker-T is used for training and validation, while VieSpeaker-H is used for testing. VieSpeaker-H is speaker-disjoint from the training and validation partitions.

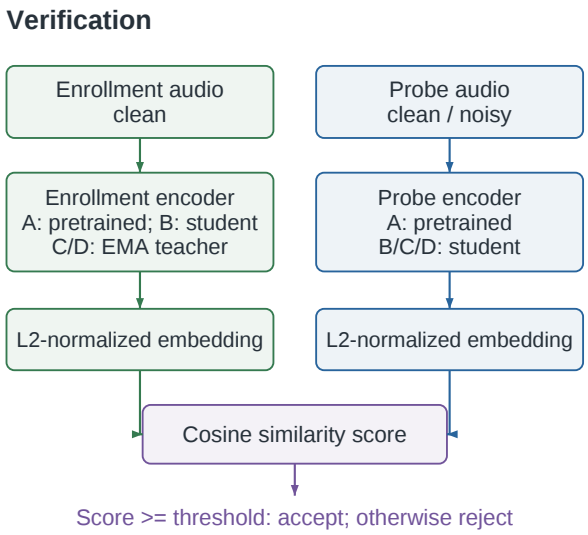


Figure 2: Verification. Encoder roles match the recorded evaluation: pretrained/pretrained for A, student/student for B, and EMA-teacher enrollment/student probes for C/D. Unit embeddings are scored by cosine similarity and thresholded.

Audio is sampled at 16 kHz. Because the audio segments are trimmed from longer recordings, we use Silero voice activity detection (VAD) to locate regions containing human speech. Recordings with insufficient usable speech are excluded from the training, validation, and testing pipelines. After preprocessing, the full-data experiment contains 3,371 training speakers and 90 validation speakers.

During evaluation, enrollment lengths range from 2 to 4 seconds, while probe lengths range from 1 to 1.5 seconds. Both are cropped from recordings in VieSpeaker-H. For target trials, the enrollment and probe belong to the same speaker but are taken from different recordings.

During training, augmentation is applied to the short student inputs. During evaluation, MUSAN noise [19] at 10 dB and 0 dB SNR is added only to probe utterances, while enrollment utterances remain unaugmented. The VieSpeaker-H trial manifest is fixed before model scoring and reused for all systems. The training, validation, and test sets are speaker-disjoint, and the test set is not used for checkpoint selection.

4.2 Training Configuration

Historical B/C/D start from the same pretrained ECAPA-TDNN checkpoint and train for 10,000 steps; A is unadapted. Each training batch contains 16 speakers, with two long–short training pairs per speaker.

The AAM margin is 0.2, and the AAM scale is 30. For Systems C and D, the EMA momentum is 0.999, and the vector-loss coefficient is 1. For System D, the Gaussian-KL coefficient is 0.01. Optimization uses AdamW [20] with an encoder learning rate of 1×10^{-5} and an AAM-head learning rate of 1×10^{-3}. Weight decay is 1×10^{-4}, and the learning-rate schedule consists of a 500-step linear warm-up followed by cosine decay. The gradient norm is clipped at 5.0, and the training random seed is 2026.

During training, the student receives short, augmented speech, while the teacher in Systems C and D receives longer, unaugmented speech. The augmentation distribution is 20% clean, 60% MUSAN, 10% room impulse response (RIR), and 10% RIR combined with MUSAN; RIR augmentation is motivated by [21].

For each adapted system, the checkpoint with the lowest noisy validation EER is selected for final evaluation. The VieSpeaker-H test results are not used for checkpoint selection.

4.3 Exploratory Pooling Variants

These variants all use AAM adaptation and student/student verification; their names do not denote objective Systems A/B/C/D. **Arch-A** (ARCH_A_ORIGINAL) retains ECAPA attentive statistics pooling. **Arch-B** (ARCH_B_GLOBAL_LOCAL) applies that pool to post-aggregation features globally and in masked, overlapping 0.5-s windows with 0.25-s hops. Valid local summaries are averaged, then a learned sigmoid scalar gate mixes global and local summaries before the existing normalization/projection. **Arch-C** (ARCH_C_RELATIVE_LENGTH_GATE) adds waveform length divided by the longest waveform in its batch to that gate. This batch-relative fraction does not encode absolute duration.

The completed architecture runs used the cleaned V2 recording manifest and 3,359 eligible training speakers, versus V1 and 3,371 in the historical objective family. Arch-A used seed 20260913 and validation every 500 steps; Arch-B/C used seed 2026 and validation every 1,000. All trained for 10,000 steps; the best noisy-development checkpoints were selected at steps 7,500 (Arch-A) and 10,000 (Arch-B/C). Historical B/C/D selected steps 8,000/8,000/7,000. Consequently, Arch-A versus Arch-B/C and cross-family comparisons do not isolate architecture changes.

4.4 Evaluation Protocol and Metrics

For evaluation, enrollment durations are 2, 3, and 4 seconds, while probe durations are 1.0, 1.25, and 1.5 seconds. Each duration pair is evaluated under clean, 10 dB, and 0 dB conditions, producing a total grid of 27 conditions.

The frozen VieSpeaker-H evaluation manifest contains 5,000 initial trials. Because some recordings do not contain enough usable speech for the required duration, the number of scored trials may decrease in longer-duration conditions. Within each experiment family and condition, models share scored trial IDs. The architecture CSV contains 101,619 scored condition-rows; the historical objective family contains 101,610. The architecture exports include one additional nontarget trial repeated in nine 2-s enrollment conditions. We retain the supplied architecture population in Table 2, rather than presenting a pooled cross-family ranking. Enrollment and probe embeddings are compared using cosine similarity.

EER is the primary metric. The saved historical paired analysis uses 1,000 percentile-bootstrap replicates, resampling enrollment-speaker prefixes and retaining paired score rows. None of its 54 auxiliary-versus-AAM 95% intervals across 27 conditions excludes zero. These intervals are unadjusted for multiplicity; one-way enrollment-speaker clustering does not capture all dependence

Table 1: Historical objective-family EER (%) at 3-s enrollment and 1.25-s probes. Bold marks the lowest point estimate within each row, not statistical significance. All values in this table use the common historical scored population.

Noise	Pretrained	AAM-only	Vector-EMA	Full
Clean	20.031	**12.700**	12.816	13.113
10 dB	24.037	**15.436**	15.694	15.746
0 dB	32.783	**23.817**	23.982	24.161

through recurring probe speakers. An interval containing zero does not establish equivalence. There is one training run per system.

Duration-dependent exclusions change target and nontarget populations, so differences across durations are descriptive. An unweighted mean of condition-level EERs is not a pooled EER. EER also does not specify a deployment threshold: a fixed false-accept operating point requires development-set calibration and held-out testing.

5 RESULTS

5.1 Overall Comparison Across 27 Conditions

Across the 27 duration–noise conditions, all systems used the same frozen trial IDs within each condition. AAM-only had the lowest point-estimate EER in 24 conditions, Vector-EMA in three, and Full in none. The unweighted mean EERs were 26.142%, 18.389%, 18.585%, and 18.725% for pretrained ECAPA, AAM-only, Vector-EMA, and Full, respectively, within the historical family. These means are descriptive because duration-dependent exclusions change the number of scored trials.

5.2 Primary Protocol

At 3-s/1.25-s, AAM-only reduced pretrained EER by 7.331 percentage points in clean speech, 8.601 points at 10 dB, and 8.966 points at 0 dB. Vector-EMA was 0.116–0.258 points worse than AAM-only, and Full was 0.310–0.413 points worse; neither teacher–student system achieved the lowest EER in a primary condition (Table 1).

5.3 Effects of Duration and Noise

For AAM-only, moving from 2-s/1-s to 4-s/1.5-s (Table 3) reduced EER from 16.990% to 10.551% in clean speech, from 20.820% to 13.616% at 10 dB, and from 28.076% to 22.801% at 0 dB. Because VAD exclusions differ by duration, these are descriptive rather than perfectly paired duration effects.

Averaged over the nine duration pairs, AAM-only EER rose from 13.416% in clean speech to 16.801% at 10 dB and 24.951% at 0 dB. Even at 4-s/1.5-s, the 0-dB EER remained 22.801%, versus 10.551% in clean speech.

5.4 Contribution of the Auxiliary Objectives

Vector-EMA beat AAM-only only in three 10-dB cases, by 0.090–0.287 percentage points, and was 0.196 percentage points worse on average. Full was worse than AAM-only in all 27 conditions, by 0.335 percentage points on average. The saved paired intervals do not establish an auxiliary gain over AAM-only.

Table 2: Exploratory architecture EER (%) for all 27 conditions on the supplied native architecture scored population. N is the scored-trial count per condition. Bold marks the exact lowest point estimate before rounding; both are bold only for an exact minimum tie. The final row is the unweighted mean of 27 condition-level EERs, not a pooled EER. Arch-A/B/C are defined in Methods and are distinct from Systems A/B/C/D.

Enroll (s)	Probe (s)	Noise	N	Arch-A	Arch-B	Arch-C
2	1	Clean	4345	17.35	**17.26**	17.53
		10 dB	4345	21.27	21.35	**20.98**
		0 dB	4345	28.55	28.50	**28.44**
2	1.25	Clean	4243	**15.54**	15.72	15.72
		10 dB	4243	18.34	**18.06**	18.34
		0 dB	4243	**25.48**	25.72	25.72
2	1.5	Clean	4132	13.49	**13.17**	13.21
		10 dB	4132	16.68	**16.62**	16.76
		0 dB	4132	**24.60**	24.87	25.30
3	1	Clean	3840	**15.36**	15.62	15.63
		10 dB	3840	19.93	19.98	**19.78**
		0 dB	3840	27.59	**27.28**	27.41
3	1.25	Clean	3755	**12.98**	13.37	13.37
		10 dB	3755	**15.90**	16.42	16.52
		0 dB	3755	24.20	**23.95**	24.21
3	1.5	Clean	3654	11.92	**11.84**	12.05
		10 dB	3654	**14.89**	15.16	14.99
		0 dB	3654	23.32	**23.26**	23.32
4	1	Clean	3377	13.87	**13.68**	**13.68**
		10 dB	3377	18.90	**18.66**	**18.66**
		0 dB	3377	**26.13**	26.33	26.67
4	1.25	Clean	3305	**12.98**	13.04	13.21
		10 dB	3305	**15.19**	15.63	15.72
		0 dB	3305	25.08	**24.21**	24.45
4	1.5	Clean	3222	**10.79**	11.14	11.14
		10 dB	3222	**14.17**	14.52	14.46
		0 dB	3222	**23.06**	23.52	23.58
Condition mean				**18.80**	18.85	18.92

Table 3: AAM-only EER (%) at two duration endpoints. Values compare different eligible trial populations across durations; each noise condition uses the same duration pair.

Enrollment / probe	Clean	10 dB	0 dB
2 s / 1.0 s	16.990	20.820	28.076
4 s / 1.5 s	10.551	13.616	22.801

5.5 Architecture Exploration

Table 2 reports the complete architecture grid, preserving the supplied scored populations and all 27 conditions. The condition-mean EERs are 18.798%, 18.848%, and 18.921% for Arch-A, Arch-B, and Arch-C. Arch-B improves on Arch-A in 12 conditions and Arch-C in nine, but neither has a lower mean. At 3-s/1.25-s and 0 dB, Arch-B reaches 23.95% versus Arch-A's 24.20%; at 10 dB, it rises

to 16.42% versus 15.90%. These completed runs show condition-specific changes without a consistent aggregate gain. Seed, selection cadence, and population differences limit attribution; the findings establish neither equivalence nor universal failure of modified pooling.

6 DISCUSSION AND LIMITATIONS

AAM adaptation supplies the dominant observed improvement. Its angular supervision supports cosine verification and adapts the encoder to Vietnamese, short, augmented speech. This comparison does not isolate supervision from augmentation or domain adaptation. Vector alignment yields only small gains in three 10-dB conditions. An EMA reference is not independent of its student, and alignment may compete with classification when crops lose discriminative information. These are possible explanations, not experimentally isolated mechanisms.

Batch-level KL matches moments across multiple identities without explicitly preserving class-conditional geometry. Its sensitivity to batch composition may help explain the observed point estimates, but the result concerns this formulation rather than distribution alignment generally. Teacher enrollment in C/D also changes inference relative to B, preventing a loss-only causal interpretation. Similarly, the pooling runs remain exploratory because training population, seed, and checkpoint cadence differ.

The main limitations are single training runs, synthetic MUSAN corruption, and duration-dependent exclusions. One-way enrollment-speaker bootstrap intervals do not capture every dependence through probe speakers; they are unadjusted for multiplicity, and containing zero does not establish equivalence. Synthetic noisy probes with clean enrollment do not cover all microphones, reverberation, competing talkers, or classroom arrangements. Longer speech is associated with lower EER, but even the longer AAM endpoint remains difficult at 0 dB.

Future work should use matched seeds, data eligibility, validation schedules, and scored rows; test speaker-conditioned alignment or a fixed teacher; and evaluate real environmental noise. An absolute-duration gate would be a new experiment, distinct from the completed batch-relative Arch-C. Application thresholds should be calibrated on development speakers at the required false-accept operating point.

7 CONCLUSION

AAM-only has the lowest EER point estimate in 24 of 27 conditions within the historical objective family. EMA/vector and batch-level Gaussian-KL objectives provide no consistent gain, and the saved paired intervals do not establish an auxiliary advantage. The separate pooling exploration likewise shows condition-specific changes without an aggregate gain. Longer speech helps descriptively, while severe noise remains difficult. Matched repeated runs and real-noise evaluation are needed to strengthen these findings.

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